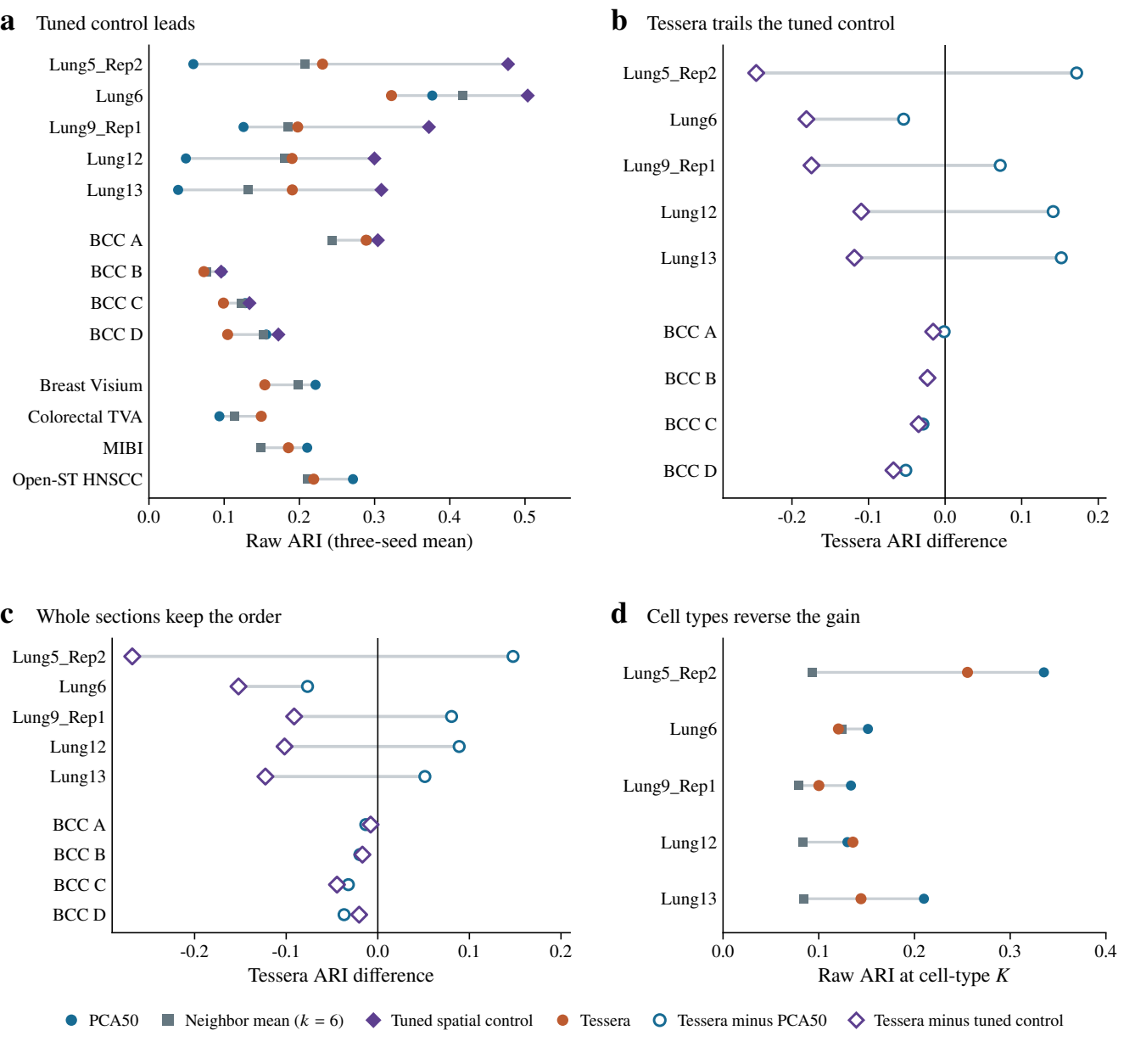

a
Tuned control leads
Lung5_Rep2
Lung6
Lung9_Rep1
Lung12
Lung13
BCC A
BCC B
BCC C
BCC D
Breast Visium
Colorectal TVA
MIBI
Open-ST HNSCC
0.0
0.1
0.2
0.3
0.4
0.5
Raw ARI (three-seed mean)
b
Tessera trails the tuned control
Lung5_Rep2
Lung6
Lung9_Rep1
Lung12
Lung13
BCC A
BCC B
BCC C
BCC D
-0.2
-0.1
0.0
0.1
0.2
Tessera ARI difference
c
Whole sections keep the order
Lung5_Rep2
Lung6
Lung9_Rep1
Lung12
Lung13
BCC A
BCC B
BCC C
BCC D
-0.2
-0.1
0.0
0.1
0.2
Tessera ARI difference
d
Cell types reverse the gain
Lung5_Rep2
Lung6
Lung9_Rep1
Lung12
Lung13
0.0
0.1
0.2
0.3
0.4
Raw ARI at cell-type K
PCA50
Neighbor mean (k = 6)
Tuned spatial control
Tessera
Tessera minus PCA50
Tessera minus tuned control